Image Processing and Medical Imaging

Discrete Tomography

This work was supported by the construction EFOP-3.4.3-16-2016-00021. The project was supported by the European Union, co-financed by the European Social Fund.

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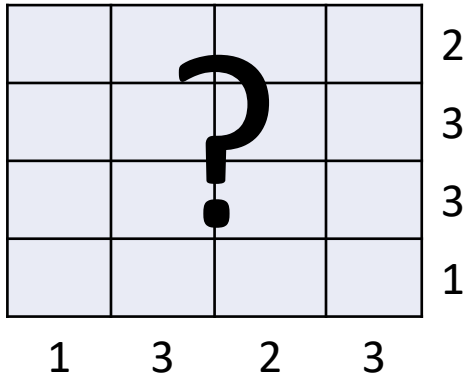
Discrete Tomography

- Discrete tomography: Reconstruction of a discrete set (matrix with integer values) from its projections (2-10).
- Binary tomography: the matrix has only binary values.
- Trivial necessary condition for existence of the solution: rowsum(9)=colsum(9)



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Examples: The reconstruction is **unique**

3			
2			
1			
	3	2	1

3	1	1	1
2	1		
1	1		
	3	2	1

3	1	1	1
2	1		0
1	1	0	0
	3	2	1

3	1	1	1
2	1	1	0
1	1	0	0
	3	2	1

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Examples: The reconstruction is **inconsistent**

3			
3			
1			
	3	3	1

3	1	1	1
3			
1			
	3	3	1

3	1	1	1
3	1	1	1
1			
	3	3	1

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Examples: The reconstruction is **non-unique**

3			
2			
2			
	3	2	2

3	1	1	1
2	1		
2	1		
	3	2	2

3	1	1	1
2	1	1	0
2	1	0	1
	3	2	2

3	1	1	1
2	1	0	1
2	1	1	0
	3	2	2

Swirching component

Uniqueness and Switching Components

1	1	1	1	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	0	0	0	0	1
1	1	0	0	0	0	0	1
1	1	0	0	0	1	0	1

Switching component

1	0
0	1

The presence of a switching component is necessary and sufficient for non-uniqueness

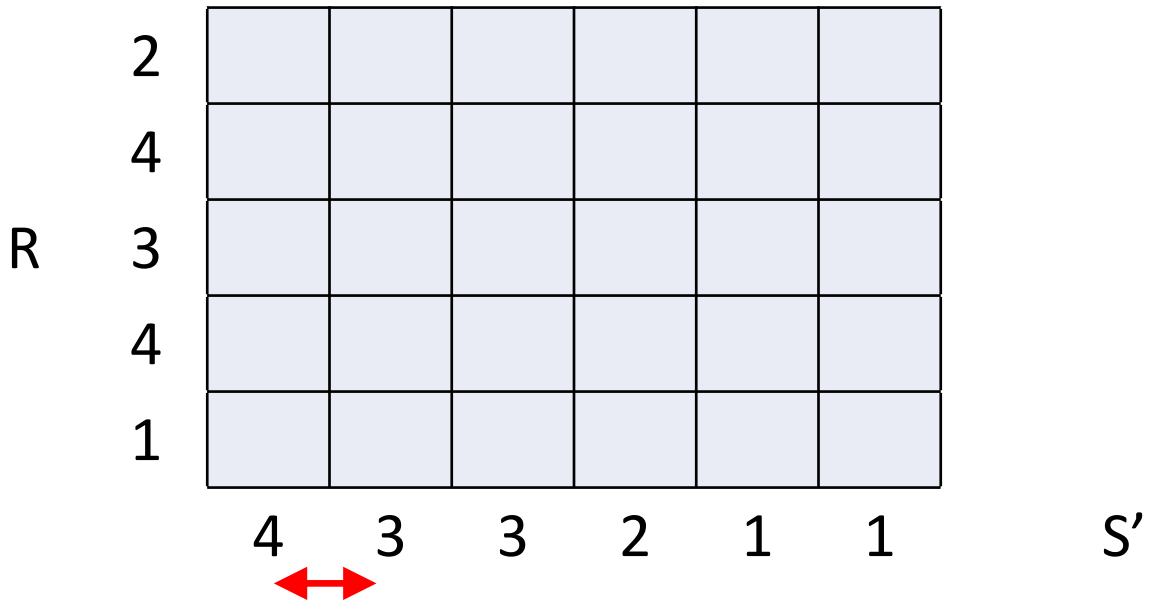
Reconstruction – Ryser, 1957

- Construct the non increasing permutation of the element of sum of the columns (denoted by *S*). The permutation is denoted by *S'.*
- Fill the rows from left to right. The result is denoted by *B*.
- Shift elements from the rightmost columns of B to the column, where S(B)<S'
- Apply the inverse of permutation that was used to construct *S'*
- Gale and Ryser (1957) gives the conditions of the reconstruction

Reconstruction – Ryser's algorithm (1)

	2						
	4						
R	3						
	4						
	1						
		3	4	3	2	1	1	S

Reconstruction – Ryser's algorithm (2)



Reconstruction – Ryser's algorithm (3)

	2	1	1					
	4	1	1	1	1			
R	3	1	1	1				=B
	4	1	1	1	1			
	1	1						
		5	4	3	2	0	0	S(B)
		4	3	3	2	1	1	S'

Reconstruction – Ryser's algorithm (4)

	2	1	1					
	4	1	1	1	1			
R	3	1	1	1				=B
	4	1	1	1	1			
	1	1						
		5	4	3	2	0	0	S(B)
		4	3	3	2	1	1	S'

Reconstruction – Ryser's algorithm (5)

	2	1	1					
	4	1	1	1			1	
R	3	1	1	1				=B
	4	1	1	1	1			
	1	1						
		5	4	3	1	0	1	S(B)
		4	3	3	2	1	1	S'

Reconstruction – Ryser's algorithm (6)

	2	1	1					
	4	1	1	1	→		1	
R	3	1	1	1	→			=B
	4	1	1	1		1		
	1	1						
		5	4	3	0	1	1	S(B)
		4	3	3	2	1	1	S'

Reconstruction – Ryser's algorithm (7)

	2	1	1					
	4	1	1		1		1	
R	3	1	1		1			=B
	4	1	1	1		1		
	1	1						
		5	4	1	2	1	1	S(B)
		4	3	3	2	1	1	S'

Reconstruction – Ryser's algorithm (8)

	2	1		1				
	4	1		1	1		1	
R	3	1	1		1			=B
	4	1	1	1		1		
	1	1						
		5	2	3	2	1	1	S(B)
		4	3	3	2	1	1	S'

Reconstruction – Ryser's algorithm (8)

	2		1	1				
	4	1		1	1		1	
R	3	1	1		1			=B
	4	1	1	1		1		
	1	1						
		4	3	3	2	1	1	S(B)
		4	3	3	2	1	1	S'

Reconstruction – Ryser's algorithm (9)

	2	1		1				
	4		1	1	1		1	
R	3	1	1		1			=B
	4	1	1	1		1		
	1		1					
		3	4	3	2	1	1	S

Convexity

- We can give better algorithm, if we have some extra conditions. For example it can be the convexity of “1”s in the matrix.
- The convexity property means that the series of 1’s is not broken:
  - Horizontal convexity (h-convexity)
  - Vertical convexity (v-convexity)
  - Horizontal and vertical convexity (hv-convexity)
  - No convexity

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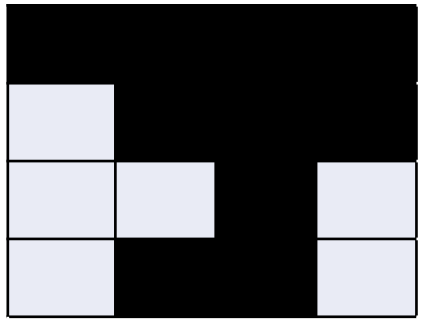
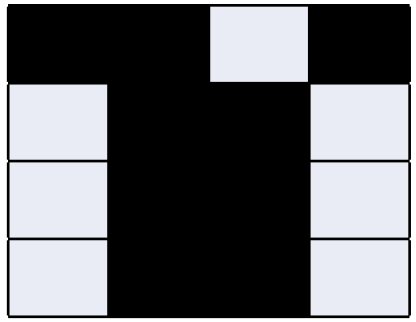
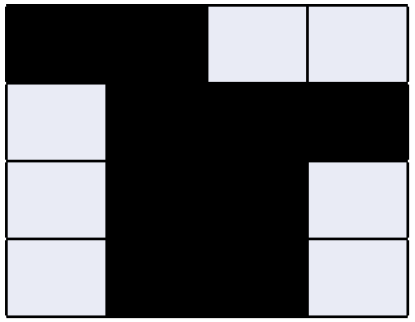
Convexity



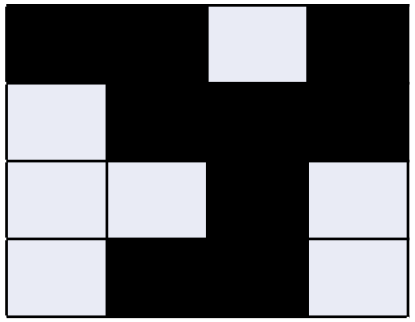
h-convex



v-convex



hv-convex



no convex

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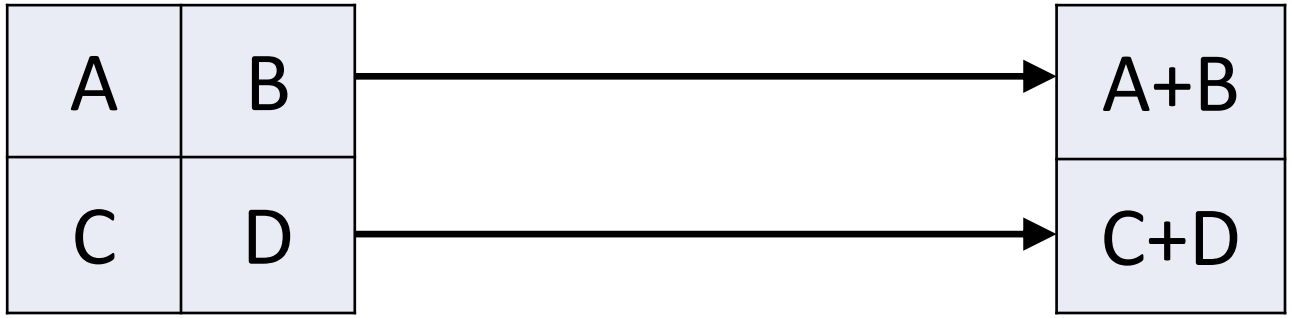
Back Projection – Matrix With Integer Values



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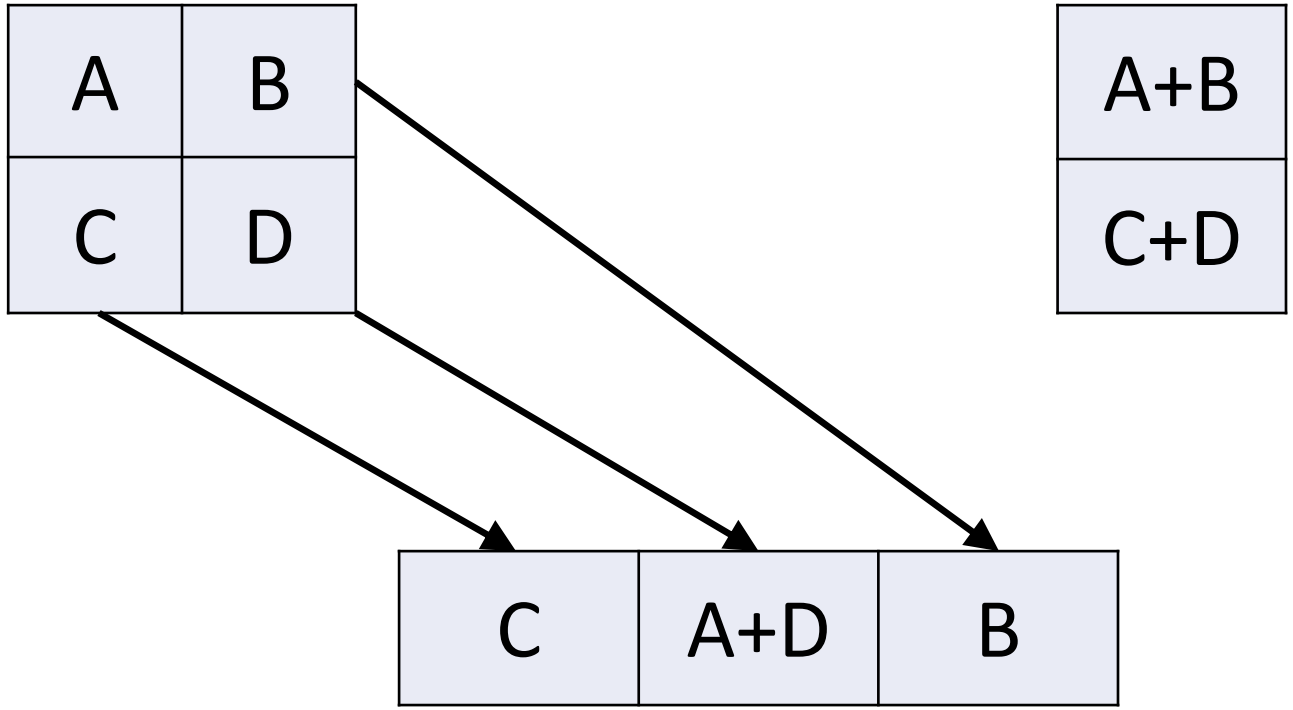
Back Projection (2)



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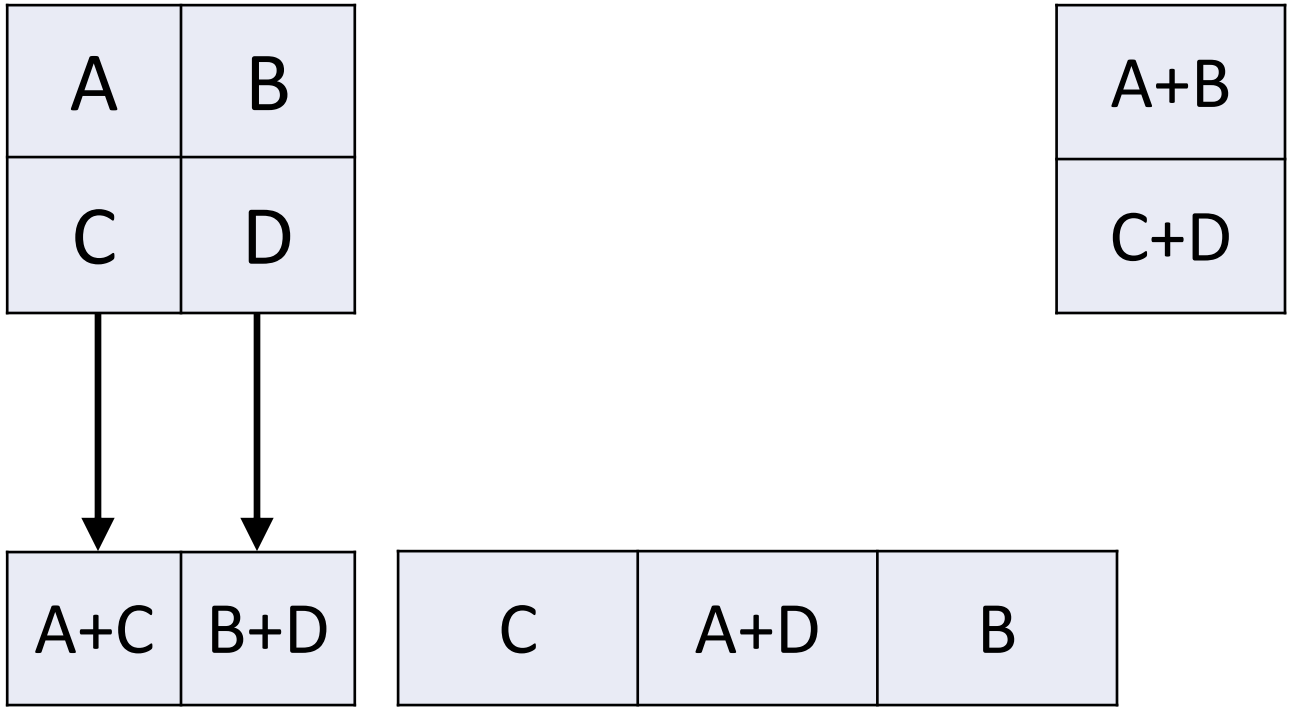
Back Projection (3)



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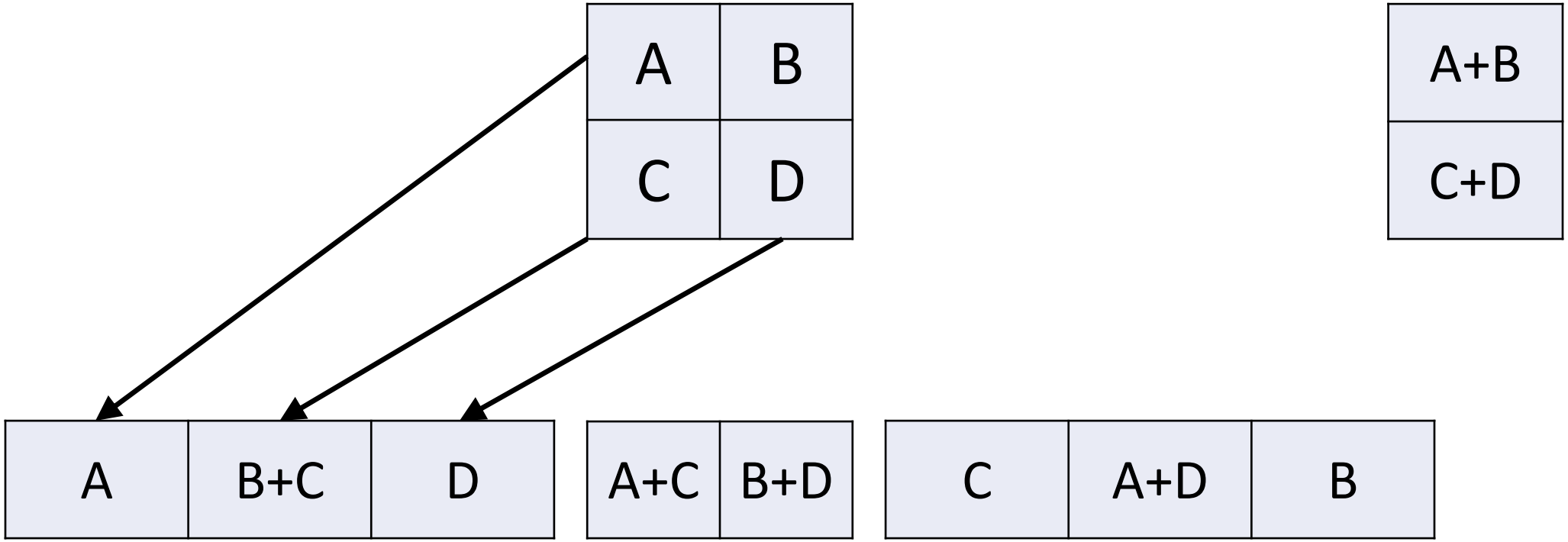
Back Projection (4)



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Back Projection (5)

A	B
C	D

A+B
C+D

A	B+C	D

A+C	B+D

C	A+D	B

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Back Projection (6)

?	?
?	?

A+B
C+D

A	B+C	D

A+C	B+D

C	A+D	B

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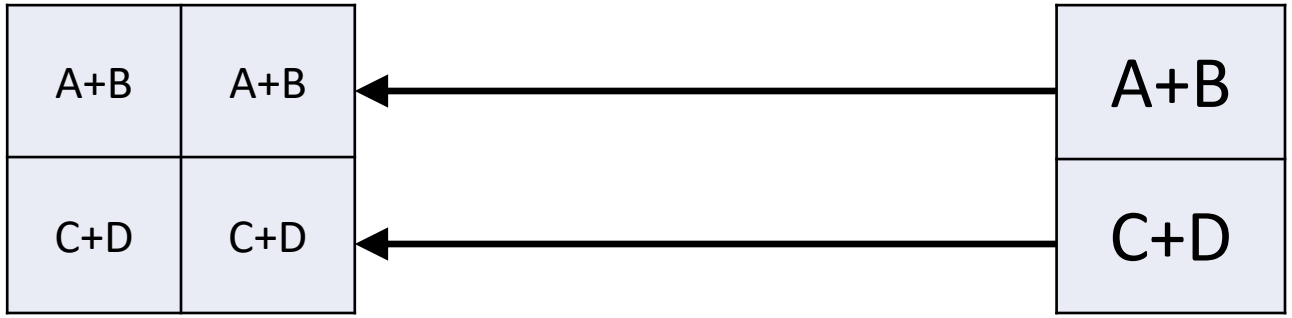
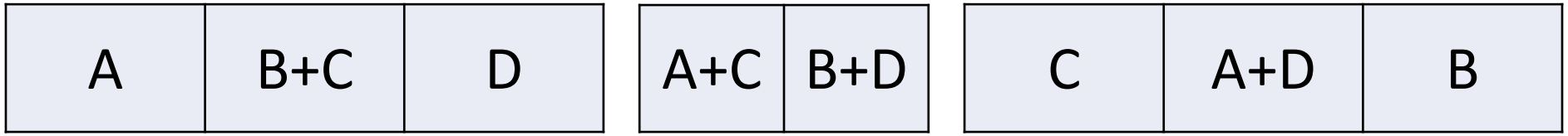
Back Projection (7)





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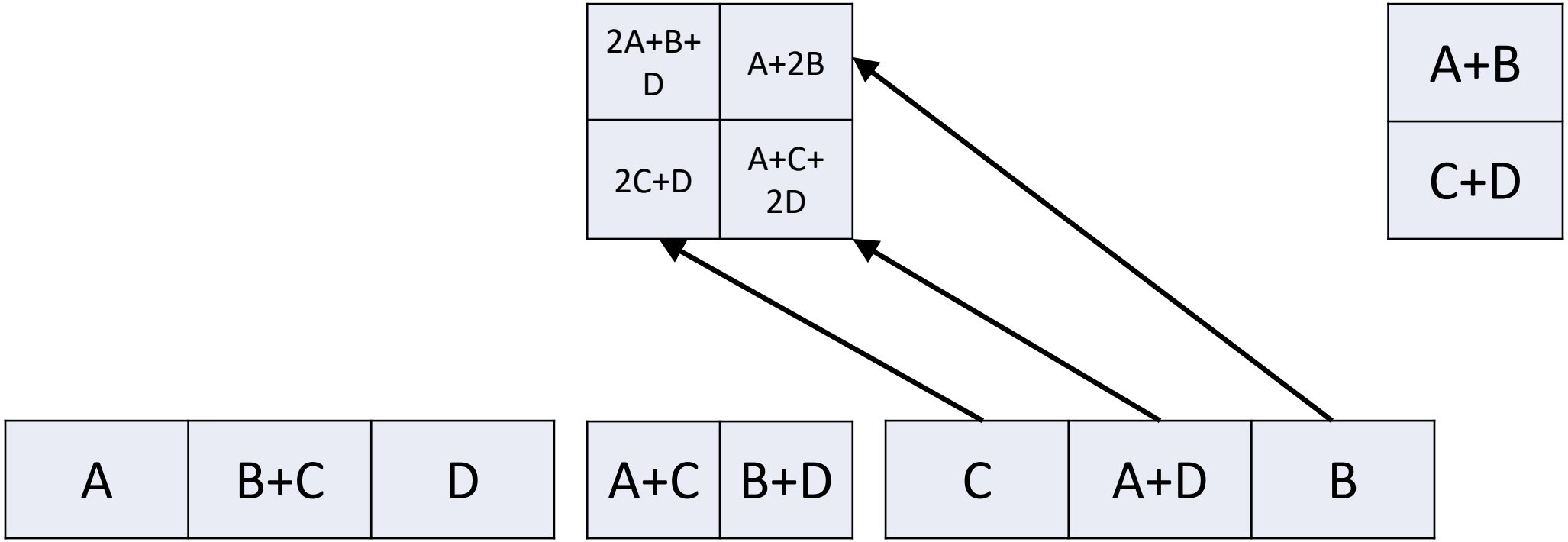
Back Projection (8)



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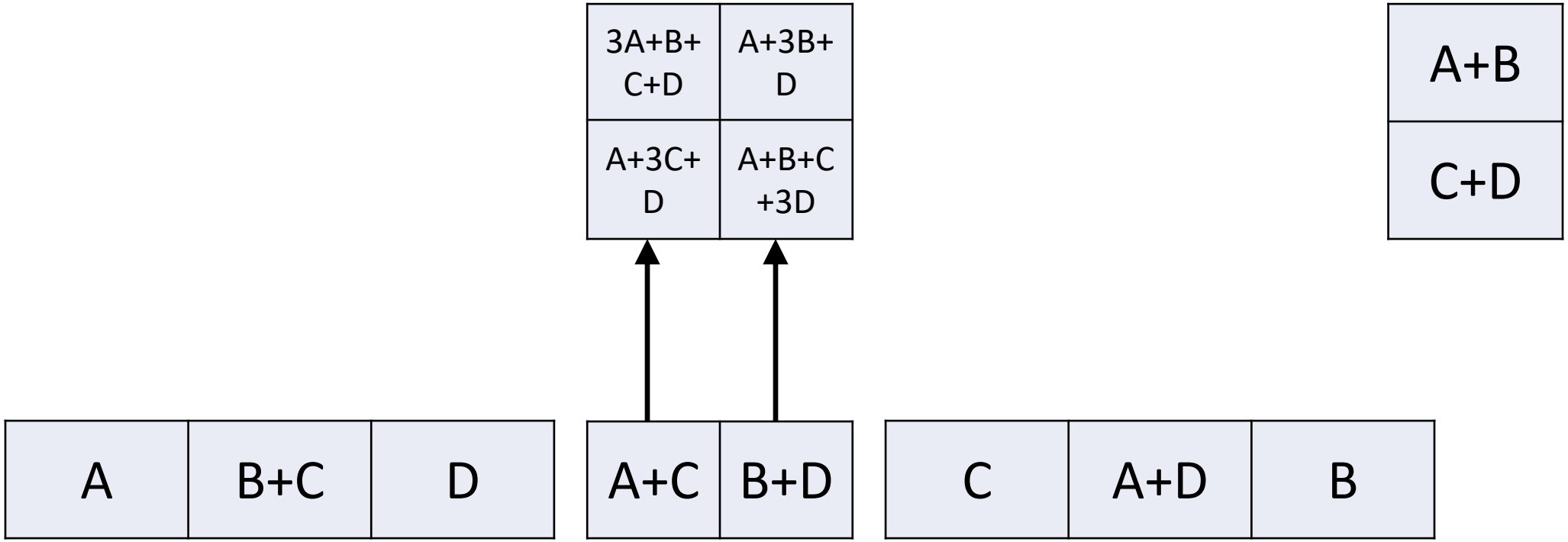
Back Projection (9)



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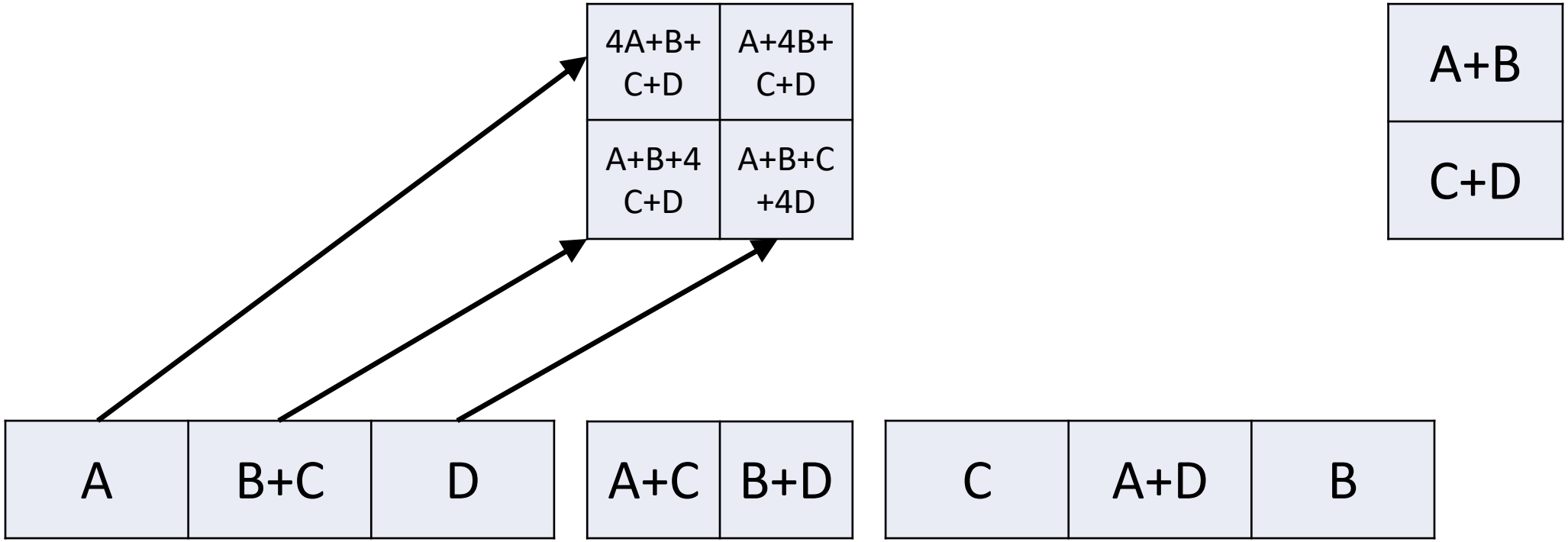
Back Projection (10)



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Back Projection (11)

3A+A+B+C+D	3B+A+B+C+D
3C+A+B+C+D	3D+A+B+C+D

A+B
C+D

Subtraction of the projections

A	B+C	D

A+C	B+D

C	A+D	B

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Back Projection (12)

3A	3B
3C	3D

A+B
C+D

A	B+C	D

A+C	B+D

C	A+D	B

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Back Projection (13)

A	B
C	D

A+B
C+D

A	B+C	D

A+C	B+D

C	A+D	B

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Thank you for your attention